Buffers Lab

AP Chemistry · Virtual Lab Sheet Name _____ Date _____ Period _____

BACKGROUND

In this experiment, we will use two different methods to prepare buffered solutions with the same assigned pH. **Buffer 1** will be prepared using household vinegar (acetic acid), $HC_2H_3O_2$, and sodium acetate, $NaC_2H_3O_2$. **Buffer 2** will be prepared using household vinegar, $HC_2H_3O_2$, and sodium hydroxide, NaOH. Both buffers will have a target pH of **4.75**. Acetic acid has a $K_a = 1.76 \times 10^{-5}$.

Objective: The purpose of this lab is to use two different methods to prepare buffered solutions with the exact same pH.

PRE-LAB QUESTIONS

PART A · PRACTICE (QUESTIONS 1–3): buffer math practice with citric acid. These are **NOT** related to the virtual lab.

1. Determine what mass of sodium citrate, $NaH_2C_6H_5O_7$, would be required to prepare a buffer, Buffer A, with a pH of 3.13 from 50.0 mL of 1.0 *M* citric acid, $H_3C_6H_5O_7$. The K_a and pK_a for citric acid are 8.4×10^{-4} and 3.13 respectively.

2. What is the predominant form of the acid in Buffer A? *Circle one:*

citric acid, $H_3C_6H_5O_7$ citrate ion, $H_2C_6H_5O_7^-$

3. A second buffered solution, Buffer B, is prepared with twice the number of moles of acid and base, but the same ratio of acid to base as Buffer A.

(a) Is the pH of Buffer B greater than, less than, or equal to 3.13? Justify your answer.

(b) Which buffered solution, Buffer A or Buffer B, would be more resistant to pH change when a strong acid or a strong base is added? Justify your answer.

PART B · FOR THE VIRTUAL LAB (QUESTIONS 4–5): these answers are the amounts you will measure out in the lab.

4. Determine the mass, in grams, of sodium acetate that should be added to 50.0 mL of 0.83 *M* $HC_2H_3O_2(aq)$ to make a buffer solution with a pH of 4.75. *(This will be used in Buffer 1, procedure step 2.)*

5. Determine the volume, in mL, of 0.50 *M* NaOH(*aq*) that should be added to 50.0 mL of 0.83 *M* $HC_2H_3O_2(aq)$ to make a buffer solution with a pH of 4.75. *(This will be used in Buffer 2, procedure step 2.)*

DATA

	Buffer 1	Buffer 2
Assigned pH	4.75	4.75
Volume of 0.83 *M* $HC_2H_3O_2$ (mL)		
Mass of $NaC_2H_3O_2$ (g)		N/A
Volume of 0.50 *M* NaOH (mL)	N/A	
Actual pH		

CALCULATIONS

1. Calculate your percent error using the assigned pH and your actual pH for both Buffer 1 and Buffer 2.

$$percent\ error = \frac{|\ actual\ pH - assigned\ pH\ |}{assigned\ pH} \times 100$$

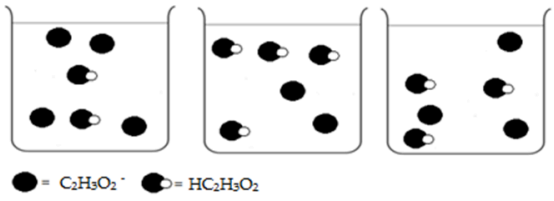
Buffer 1	**Buffer 2**

ANALYSIS

2. If either percent error is greater than 5%, provide a reasonable source of error and explain precisely how the error would affect the actual pH.

3. A different lab group attempts to make a buffer with an acid and a salt. They choose HCl and NaCl. Explain why their attempt will not produce a solution that is resistant to pH change. Justify your answer using net-ionic equations.

4. Circle the beaker below that best represents the particulate diagram for Buffer 1. Explain your reasoning referencing the Henderson-Hasselbalch equation.



5. It is observed that Buffer 2 changes more significantly than Buffer 1 upon the addition of strong acid. Explain this observation.